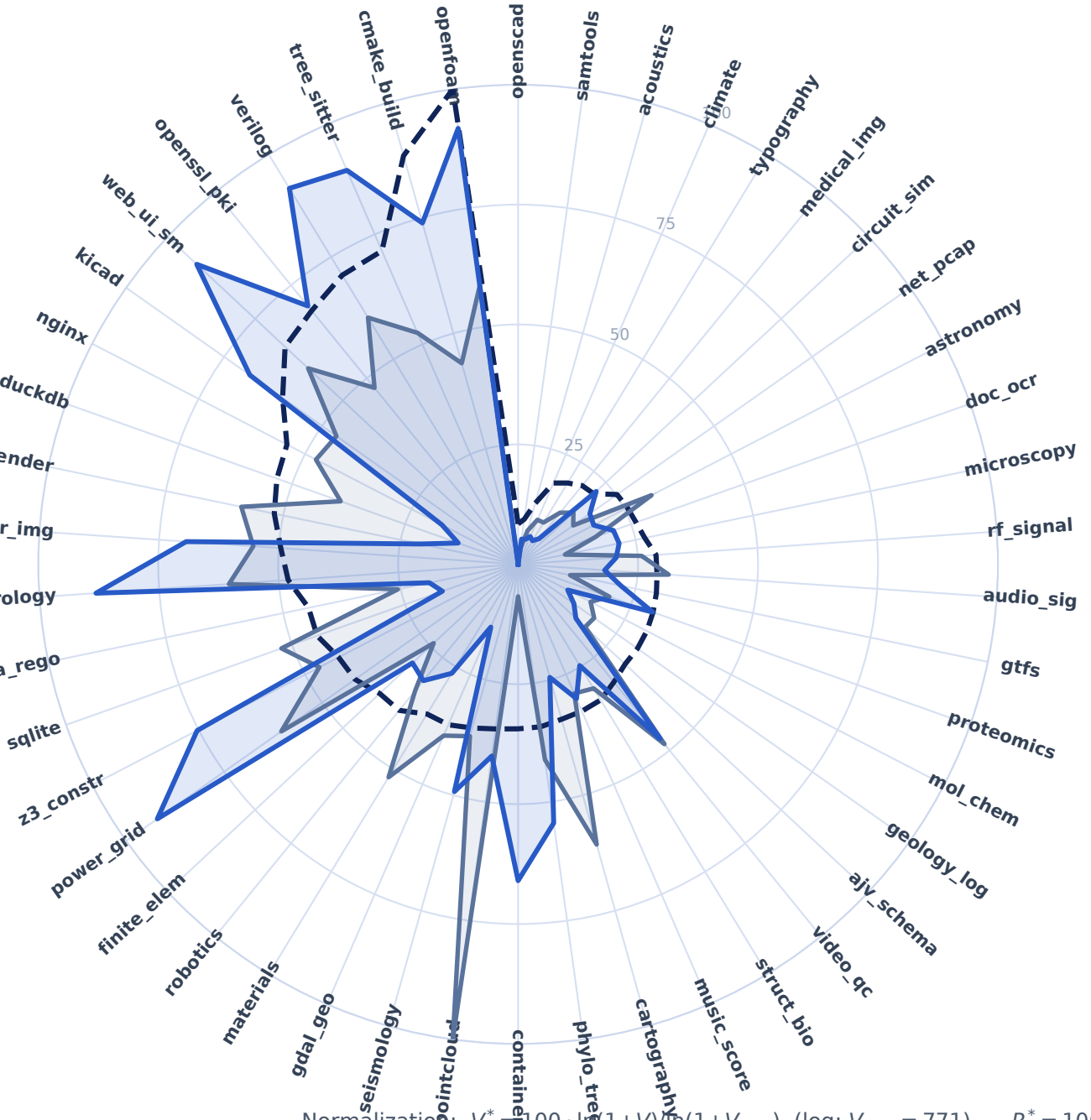
Per-Family Radar — Validator Count vs. Run 1 & Run 2 Reward

46 task families (ordered by validator count) · Qwen3-27B Harbor PPO



Validator count V^*

Run 1 reward R_1^*

Run 2 reward R_2^*

How to read

- The dashed navy outline is each family's validator count.
- Where a reward series (grey = Run 1, blue = Run 2) bulges OUT toward the validator outline, that family's reward rides its validator count.
- Where it stays small while the validator outline is large, reward is decoupled from validator quantity.

Run 2's blue shape is smoother across spokes and detaches from the navy outline on the high-validator families (cmake, openfoam, tree_sitter), i.e. the success-smoothed reward no longer inflates with validator count.

Normalization: $V^* = 100 \cdot \ln(1+V)/\ln(1+V_{max})$ (log; $V_{max} = 771$) $R^* = 100 \cdot R/R_{max}$ (shared $R_{max} = 1.63$ over both runs)